

ActInf Textbook Group ~ Cohort 3 ~ Meeting 18 (Chapter 8, part 1)

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All right, welcome everyone.

It's 7-11-23.

We're in cohort three in our first discussion of chapter eight.

So we'll head over to chapter eight and look at some questions and comments.

But first, does anyone want to bring up any just general thoughts or reflections on chapter eight?

Ali?

Well, yes.

Actually, chapter eight, as obviously the title suggests, is kind of complementary to chapter seven in terms of being the continuous time counterpart to the active inference formulation of discrete time.

So as we saw previously in chapter four, we have both a continuous time version and discrete time version of active inference employing different mathematical technologies and techniques.

But of course, they can be regarded as kind of formulating the same or at least similar formulation just using different mathematical techniques.

So namely, in discrete time case, we use matrices to represent states, transitions and so on.

But in continuous time, we use generalized coordinates in order to describe the states and not just, I mean, discrete matrices for states.

And also, the starting point of formalizing active inference in continuous time also differs a bit.

Because in continuous time version, we begin with a version or a variant of Eido's stochastic equation.

So we're going to be able to do a version by representing a flow plus random variable and then trying to somehow formulate the flows in terms of increasingly granular way.

And yeah, tracing the trajectory through the state space, continuous time state space, employing that, I mean, Taylor expansions and other mathematical technologies and so on.

Great. Olivier?

Yeah.

What do you mean by granular here?

So, I mean, Eido's stochastic equation is a very general equation describing any kind of, sorry, differential stochastic phenomenon.

But in this case, we focus on a particular kinds of state space with particular kinds of behaviors.

So, especially in active inference, we focus on active agents as opposed to a more general sense of the term, such as, I mean, describing the whole behavior

behavior of self-organizing system or even a simple system that can be basically described with a simple statistical physics.

So, the focus of active inference, as I said, is the agents that don't have empty set of active states, right?

But of course, it can also be generalized to other kinds of agents with empty active states.

But the active inference framework or FEP don't have anything particularly interesting to say about those

other kinds of agents.

That's what I meant of granular.

We focus on a subset of the systems that can be described through Eido's stochastic equation.

Okay.

Cool.

Thank you.

We can come back to some of these.

And also, Ali, we will be keeping in mind that we need to make the videos for 7, 8, 9, 4, 5.

Sure.

Yeah.

Whatever you want.

Okay.

Anyone else want to just have any general thought or question, raising their hand or in the chat on 8 and continuous time?

I really like that chapters 2 and 3 give us the kind of two ways to get there, to act imph, the low road and the high road, the how and the why.

And now it's like we're in the city and now there's two restaurants or there's like two doors to the restaurant with the discrete and the continuous time.

So it's another dyad of chapters with 7 and 8 and another juxtaposition that gives us parallax and helps us understand more about each of these two methods by highlighting, on one hand, situations where they seem to apply more naturally.

For example, the discrete time formulation seems to be a natural fit for discrete decision-making tasks, like which slot machine are you going to pull?

Whereas the continuous time model seems to be a more natural fit for analog perception control systems, like proprioception and sensory motor behavior.

And chapter 8 is also going to get us to hybrid nested models.

For example, a model with two layers where the lower level is continuous time and the higher level is discrete time.

So that kind of represents like a decision-making central nervous system that's dispatching commands to a continuous time sensory motor body.

And that is also explored in the sort of folk psychological active inference work where you can kind of get the synthesis of continuous and discrete time models.

Anyone else want to add a comment on 8 before we start to look through it?

Yeah, I have a question.

Wait, I'll just...

Maybe it's nicer if I have my video on.

Let's see.

I was wondering if there are any good resources on applications of continuous active inference.

Because up until now, I've worked on two projects with active inference, but only working with discrete

state-based models.

And I was wondering if there is any like good papers on continuous active inference.

And it's sort of representative applications.

Are there in the situation?

Yeah.

I think this table has some...

Oh, yeah.

Yeah, I'm so sorry for asking a redundant question.

No, no, it's all good.

These are not all the applications, but let's just look at the topics that they have raised.

So the bird song is definitely a classic.

Single singing bird and two singing birds.

Okay.

So the bird song is...

Ocular motor behavior.

Okay.

Reflex behavior.

More eye behavior.

Different cognitive states.

More eye behavior.

This is one of the few papers on mirror neurons.

Okay.

Attention.

Attention.

I believe this is attention being modeled in a slightly different approach.

This is a continuous time attentional framework, whereas the Sanved Smith paper uses discrete time attention.

Hybrid and self-organizing.

But there's other citations, and I'm sure you could come up with many more things.

And also, like, one of the kind of model annotation tasks that I think would be super helpful in terms of meta-analysis would be, like, for each paper,

how can we describe the active inference models they used?

So then we could just, like, filter a table.

Where are the discrete time models?

Where are the continuous time models?

Where are the nested models?

Where are the models that use empirical data?

Yeah.

Yeah.

Yeah.

Thank you for the overview.

Cool.

Okay.

Okay.

So, from the figures,

figure 4.3,

recall that the discrete time and the continuous time generative models have some structural similarities.

At the same time, they have important differences.

And also, partially, this is signaled with a difference in notation.

So, like, hidden states are going to be reflected with X in continuous time, not S .

Observables are reflected with Y , not O .

Causal interventions in the system are reflected in V , not π .

And really, the most important difference is that, whereas in the discrete time, you have past, present, and future explicit state predictions,

in the continuous time setting, you're using the generalized coordinates of motion, where X is the hidden state at a given time, and $\dot{X}$ is the derivative of the hidden state, how it's changing through time.

So, the B matrix, which is the transition matrix in discrete time, describing how states go from one time step to another, how the hidden states change, here is describing, like, kind of like the derivative or the integration function, because it's describing how derivatives are linked to each other.

And what causes do in the model is they influence how derivatives change through time.

Here's the Edo calculus type equations that Ali spoke to first.

First, we have observable states are a function, G of hidden states, that's the flow, and some stochastic noise term about Y , sensor noise.

And then, $\dot{X}$, rate of change in X , is a function of the hidden state at the current time, and the causal forces on the system at that time, plus a fluctuation term on X .

Action is absent in equation 8.1.

This is because action is part of the generative process, not the generative model.

Pending our definitions of generative process and generative model.

The generative model only deals with those variables directly influenced by states external to a Markov blanket.

What do you think about this, Ali?

Ali, I think it's a function of the general model.

Well, yeah, actually, this is one of the points that was kind of brought up in the discussion with Maxwell.

Well, because as we know, generative model, I mean, if we consider FEP as this kind of general framework or theoretical framework to describe the behavior of the things, quote unquote, things that persist through time,

we can theoretically describe the generative model as containing any kinds of different situations, both

containing activist states and or in other words, with activist states and also with empty set of active states.

So, but as I said before, I believe active inference is more concerned with the situations that have non-empty set of active states.

In other words, they have the ability to perform or to undertake or to put it in more rigorous terms to kind of engage in a causal intervention with the environment,

both in terms of updating the generative model and also effectuating some changes in the environment that can fit better with their generative model.

So, yeah, that's, I mean, what's meant here by directly influenced by states external to a Markov blanket means the states from the environment that can dynamically act upon the Markov blanket

in order to make it in order to make it more suitable or more suitable or more adapt to the environment through the generation of the Markov blanket.

I'm sorry, the generative model and also minimizing the free energy.

Yeah, thank you.

It looks like the difference here between 8.1 and 8.2, other than the bolding of the equations, is like V , a static variable, is replaced by a controllable action parameter.

So, now a specific implementation of equations 8.1 and 8.2 are going to be described.

So, now, Ali, go for it.

Sorry, I forgot to mention something.

There was a debate on Twitter a couple of days ago about whether or not FEP can be applied to inert objects such as rocks,

which is, I mean, one of the debates that's been going on for a long time now.

But, you see, theoretically, it can be applied to any kinds of object, even to an inert rock.

But the only difference here is that inert rock has an empty set of active estates.

That's it.

So, if we take a non-empty set of active estates, such as, I don't know, biological organisms or sentient agents and so on,

that would also employ the same kind of mathematical technologies.

But the only difference would be the existence of non-empty set of active estates.

Sure.

I would almost also say, though, that it's not that the active states are empty for an inert or a classical or a conservative object,

that they're just trivial.

For example, just outputting a stationary action state.

Okay.

So, now they're going to give a specific example of what this pair of equations looks like.

And there's a point attractor.

So, this is going to be kind of moving us towards a continuous time system where there's a fixed attracting point,

where if the actuator is below the fixed point, it's going to go up.

And if it's above the fixed point, it's going to go down.

And so, this is going to be the kind of model that's used for eye saccades and for, like, elbow movement and things like that.

So, generative process is specified.

The hidden state is basically just passed through, and the action is basically just passed through.

V is in the generative model.

V.

U is in the generative process.

V.

This connects to the equilibrium point hypothesis.

Again, it relates to a proprioceptive prediction about expected proprioceptive feedback coming from, for example, the limbs or the eyes.

V.

This enables action to be prediction fulfillment, rather than, for example, reward maximization or utility maximization directly.

V.

This relates to an extensive discussion on forwards and inverse modeling and interactionism and instructionism.

And some of that was explored in the skillful performance livestream.

V.

Now they're going to move from this fixed point attractor into a spring attractor.

V.

So, now there's going to be an oscillatory attraction with dampening.

V.

V.

And that is going to require not just a scalar description of the attractive point, but we're going to start moving into this generalized coordinate way of thinking, where we're going to be tracking kind of a vector of position and velocity.

V.

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! V.

V.

for continuous time discussions and stochastic differential equations and all of this,
some topics that are certainly relevant for the discrete time setting,
but are more naturally addressed in continuous time.

So in chapter five, we looked at this spinal cord cross section,
and it was discussed as a simple proprioceptive input, descending fixed point prediction,
and then the residual was being passed on, the error signal was being passed forward to engage bodily
movement.

So like if the proprioceptive data and the descending prediction were identical,
there's no discrepancy, no motion occurs.

Whereas when there is a discrepancy between the expectation of proprioception and the incoming
proprioception,

this triggers a discrepancy, and that discrepancy is resolved through movement.

So now that example from chapter five is being re-approached using the formalisms
and the continuous time setting that we see in chapter eight.

Here we have a box on precision, attention, and sensory attenuation.

This is going to cover several different topics.

First, precision and attention are described as synonymous.

That's a pretty general claim.

However, one can think of as something that's being attended to highly is giving precise inputs
and therefore is salient, it's relevant, it updates the model a lot.

If something is not being attended to in the limit,

that is equivalent to something that has zero precision,

it's infinitely ambiguous, and therefore it doesn't update the model because it's not salient, it's not relevant.

So it's kind of a general point about precision and attention.

And one way that this plays out in relationship to motor control

and some extremely interesting and unique explanations and predictions arising

about precision attention and motor control in a predictive processing setting

is this topic of sensory attenuation in motor generation.

So attenuation means diminishment.

And in some ways, attenuation, even though it sounds like basically the same word, is like the opposite of attention.

It just means it's like dis-attention or like alleviation of attention.

Does anyone want to like give a thought or explanation about what is the role of sensory attenuation in motor initiation and fulfillment?

Okay, I'll just give a thought on it.

Okay, so if one is sitting down in a chair, the accurate belief to have is that I am sitting down.

How, without invoking goals or explicit predictions about future time points,

how do we go from having an accurate belief I am sitting down to having an accurate belief that I am standing up?

I mean, the transition between those two steps, basically two things have to happen.

You have to change your mind, thinking you're sitting down to thinking that you're standing up, and you have to change the world.

You actually have to go from sitting down to standing up.

That's action.

And so how does that happen?

Well, the issue is that if you're sitting down and you have a high precision, accurate belief, I am sitting down, and then you change or you try to change your expectation or your preference or your fixed point or your set point to standing up, it's going to be like, no, that's unlikely.

All of our proprioceptive evidence is telling us that we're sitting down.

And so in that way, like the body would kind of be like an anchor, and it would pull the cognitive belief back to, no, actually, we're sitting down.

So if you just try to be like, we're sitting down, and we know that we're sitting down, and you just try to say, okay, now we're standing up, but the body's still sitting down.

If you have high attention on the proprioceptive inputs, then that cognitive belief gets dragged back.

And so you end up not initiating motor movements.

And so the resolution of this challenge is you update your belief,

I am standing up while you're sitting down.

Meanwhile, you engage in sensory attenuation.

You release the attention on the proprioceptive input.

And during that alleviation of attention to proprioceptive input,

you can think about the distribution on proprioceptive inputs to actually include standing up.

So now that is a more viable hypothesis.

The posterior probability becomes higher.

Then action selection can be engaged to fulfill that expectation about standing up.
And so voila, now you're standing up, and you think you're standing up.
And then you can re-engage precision so that now you can have that precise belief that you're standing up, whereas previously you had a precise belief that you were sitting down.
And the key step was kind of like you're driving on the road.
It's like attention is like the gas.
You pull off the gas, make the turn, and then re-engage the gas.
And another example of this is in our eye saccades.
When our gaze is fixed, we have high precision on our visual input.
When we do an eye saccade, transiently, it suppresses visual precision.
And that's why during saccades, which happens several times a second, we don't see like a big blur because it's almost like a blink.
Now, it's not your eyelids coming over your eyes, but it's actually an attentional blink when you do a saccade, which is why the highly dynamic input during a saccade is suppressed or not attended to so that we end up having the generative experience of fixed high precision vision.
And then we blink into another fixed high precision visual input with a suppression of attention during the saccade.
And different pathologies can be understood as challenges associated with initiating movements or with terminating movement or with continuing movement.
And so different pathologies have been associated with this.
So certain motor dyskinesias where somebody may have a challenge initiating a movement, but once they are moving, like they walk normally.
Or on the other hand, there are different motor pathologies.
Any thoughts or comments on this?
This is kind of an interesting topic.
Even just outside of this embodied or technical setting, just like this idea that in order for the possible to be actualized, there has to be a relaxation on the actual to enable the movement to a different space.
And here's the cyclical precision.
And this has been associated heavily with differences in the neurophysiology of vision.
Okay.
Okay.
Again, just raise your hand or write anything in the chat if you want.
A classic dynamical systems model from ecology is going to be introduced.
And this is the Laca-Volterra dynamics.
Sometimes it's called predator-prey dynamics.
Or especially in the context of neural Darwinism and dynamical systems and neuroscience,

sometimes this is called winnerless competition.

Because as you can see here in this ecological example,

it's not that the herbivores are trying to defeat the plants.

In fact, they have a relationship where if the herbivores are transiently succeeding in decreasing the plant numbers,

then they will have this demographic collapse.

And so this is the winnerless competition is you have dynamical oscillations.

And so this describes ecological dynamics, but also it describes activity patterns amongst brain regions,

like two systems that have an oscillatory repressive consequence on each other,

a repressilator that will have this kind of winnerless competition dynamic.

So this is just a classic dynamical system.

And in the SPM textbook, the Laca-Volterra is used to describe these winnerless competitions,

neural Darwinist dynamics.

These three traces, plant, herbivore, and carnivore,

you can think of them as like in this graph, like a plot happening through time.

Or you could imagine a cube with like the X, the Y, and the Z axis being plant, herbivore, carnivore dynamics.

And so this system can be represented as three traces along the X axis here.

Or you can imagine it as like a movement in a three-dimensional cube.

And so if it was spending a lot of time up in this corner,

you can think about what that would mean and so on.

And then on that cube, just like the Gödel-Escher-Bach book cover,

you can project from the three-dimensional cube down to a two-dimension.

And that would be like putting a camera on one side of the cube and just taking a picture.

So here we're looking at the herbivore by plant projection of the cube.

And it has this kind of oscillatory cycle.

Here it's approximately periodic because there's this dampening,

but you could design a system that has different behaviors.

Now the lock of Volterra is going to be applied

in a few different active inference contexts.

The first is in the eye blink conditioning.

And so this is conditioned stimulus and unconditioned stimulus.

So this is kind of like a classical Skinnerian behaviorist paradigm

of behavior action blinking associated with conditioned and unconditioned stimuli

using this dynamical systems framework.

Whereas in the discrete time, we'd have, okay, time step one,

do we get this stimulus or that stimulus or neither?

And then what do we do?

And how do we make the decision?

And then the next time step, this happens.

All of that is happening in this kind of continuous unfolding winnerless competition setting.

So that's the first example.

On the left side, that's the blink.

The unconditioned stimulus produces a blink that's alert association.

And the right example, which also listened to Thomas Parr's bookstream 2.1,
when he gives a little more of the history.

Or we have a few in the Thomas Parr 227 page.

He gave some great history here.

Talking about how the generalized filtering and continuous state space approach
was early on being the most developed.

People started wanting to get sequential dynamics.

So they developed the kind of model that we're going to see right here.

Like here, this is like sequential writing dynamics.

Clearly, there's a repetitive motif that's being repeated sequentially.

But then the desire to explicitly sequentially encode beliefs and to engage in explicit planning as inference
motivated the work in discrete state space models where planning as inference is a more natural fit.

Of course, by the end of the chapter, we're going to be kind of merging these two different kinds of models
together.

But here's an example of Locke of Volterra Dynamical Systems Dynamics in a handwriting setting.

So it's kind of like a sequence of points that are being targeted and driven through sequentially by hand.

Short box 8.2 on learning in continuous models.

Short box 8.2 on learning in continuous models.

Slightly technical note.

Perhaps Sanjeev's textbook will clarify some of these pieces.

However, using Laplace approximation, so a quadratic approximator,

we're able to use integrals on continuous distributions to specify the attention-weighted evidence gradients.

But this is just kind of a little encapsulated box.

Ali.

I just wanted to mention a small typo in box 8.2.

So the third line in the second block of equations should be minus $ds \mu \theta$.

So yeah, the minus in front of d , I suppose, should be added.

Thank you.

Good.

Glad that we have this one.

Thank you.

All right.

Now we're going to move into a second classic.

So just like the first, we saw like the simple point attractor and then the Hooke's Law attractor spring.

Now in this dynamical setting, we're going to have kind of two levels of sophistication of dynamical systems models.

The first one was the generalized locked Volterra system.

The second one is the Lorenz system, Lorenz attractor, which is a canonical example of a complex or of a chaotic system.

And it's a super fascinating story with Lorenz and his work in weather prediction.

And that was, of course, one of the most important moments of the complexity and the computational chaos research of the last hundred years.

This is also explored a lot more in live stream 34 on Markov blankets and stochastic chaos, I believe, where solitary and coupled Lorenz attractors are modeled.

It's a super fascinating work.

Um, generalized synchrony is going to be unpacked in the setting of bird song, which was a Fristin and Frith paper, maybe one slightly earlier than this, but the 2015 is kind of the classic citation.

And this is seen as a kind of exemplar of multi-agent inference problems when they are, quote, singing from the same hymn sheet, as Carl always says.

And so the bird song attractor is going to leverage the Lorenz attractor described above.

And there's going to be two Lorenz attractors coupled through communication reflecting bird song.

And some details of those attractors are going to be described and simulated.

And the kind of, um, big takeaway here is that the, um, generalized synchrony is accomplished through communication.

So here we have the first and the second bird on the X and on the Y axis.

We can think about this line, $Y = X$ line, as being a manifold that would reflect perfect synchrony.

So if you could describe movement on this two-dimensional plane by appealing only to movement along a single line, you would have reduced a two-dimensional setting into a one-dimensional setting.

And that's like a dimensional reduction.

And that's called a statistical manifold.

Before learning, while there is overall, um, adherence towards this $Y = X$ line, we see that after learning the manifold, um, the behavior is tightened immensely around the manifold.

That represents, um, the, the learning to synchronize between these two birds.

The two birds' joint trajectories confined to an almost one-dimensional subspace indicating synchronization.

Also, it's kind of a cool nod to fractal dimensions.

Like if it was perfectly on this line, you'd only need one dimension.

If it was purely covering the whole 2D space, you would need two dimensions.

So then you can think about this as being slightly higher than one-dimensional.

This connects back to the earlier attention and attenuation.

And this alternation of listening and singing.

It's kind of like, now this is not all communicative settings and there's more work on communications, but it's kind of a fun paradigm.

We're basically like the bird expects there to be singing.

Both birds expect there to be singing.

It's like, you know, insert your favorite cultural singing moment.

Um, if you're going to sing, my preferences will be fulfilled through perception.

And if I'm going to sing, my preferences are going to be fulfilled through action.

Fulfilling my preferences through action is going to reduce my sensory attention to perception.

Until eventually it's like, well, I'm not sure what's happening.

I'm going to stop singing.

And then in that silence, the other birds are going to go, oh, my preferences were being fulfilled by listening attentively.

I'm not hearing it anymore.

I better reduce this discrepancy by taking action because listening harder isn't going to help because that's not going to make the song happen.

And so that's singing from the same hymn sheet, like Carl says.

Um, and it relates to, um, joint, uh, you know, shared, shared regimes of expectation and joint perception action and multi-agent systems.

And the generalized synchrony doesn't mean lockstep.

Generalized synchrony doesn't mean both birds are singing identically and then they're both silent.

Generalized synchrony means that there's mutual information between these two coupled systems such that there is an information geometric manifold that represents their joint movement.

But it doesn't have to be a lockstep same behavior at the same time.

Turn taking is also an example of generalized synchrony.

8.5 gets to hybrid, discrete, and continuous models.

So here we see a nested model.

There's a line that's supposed to be between pi and g, obviously.

And in the top level, we see the discrete time generative model.

$T - 1$, T , and $T + 1$.

We see all the letters that we know and love from chapter 7.

A, B, D.

But then within each time step of the higher order model, we see the continuous time model with the generalized coordinates of motion.

And so this was unpacked in Livestream 46.

Active inference does not contradict folk psychology.

Where the top level was called decision active inference or DAI.

And the bottom was called motor active inference or MAI.

Because somebody can think about the discrete decision making on the top can be like, am I going to raise my left hand or raise my right hand?

And that's a discrete decision.

There's no need to make it an analog.

But then with the descending command that it's going to be, okay, raise my right hand.

Then we can engage that whole motor fixed point apparatus of the continuous time model.

Whether we do it with a lock of Altera or a Lorenzo tractor or the Hooke's law or just a simple point attractor.

So all the kind of continuous time tools that we saw earlier in the chapter, we can deploy in the setting of a nested model with a discrete time decision maker on top and a continuous time on the bottom.

Which may articulate better with the seemingly analog nature of perception and action in the last mile.

This is a little bit of machinery that helps map between the discrete selection of target locations into the continuous setting where those target locations can be realized.

And an example of this is using an isocate where there's a discrete movement of a target location followed by a continuous time trajectory of the eye circling to that.

So box 8.3, again, raises just some sort of super curt analysis of a sophisticated topic, which is a Gaussian mixture model.

And like, we know the Gaussian, we love the Gaussian, we know we contractively optimize it, and so on and so on.

It does an incredible job of approximating distributions that have a central mass tendency.

A lot of statistical tools work incredibly for Gaussian.

But of course, not all Gaussians are going to describe given complex distributions.

But the good news, whether you approach this from a Gaussian mixture model or a hierarchical Gaussian filter or just generalized filtering, is that it turns out that through composition of Gaussians, you can build complex distributions that have almost arbitrary characteristics.

So that's kind of a way to get the tractability of Gaussians, but also, for example, make a bimodal distribution or trimodal distribution.

This engages the challenge of Bayesian model reduction, Bayesian model selection, structure learning.

So that's kind of a way to get the right.

So that's kind of a way to get the right.

So that's kind of a way to get the right.

Table 8.1 closes with some discussion of continuous time model advances.

And that ends the relatively short but pretty dense chapter 8.

Does anyone, yeah, Olivier?

Yeah.

Yeah.

I'm not sure I can formulate it well, but the thing is, you know, when we think in terms of discrete time model, we usually, you know, we think in terms of regular time steps.

And with this mixed, you know, continuous and discrete time model, you know, is that possible to model a behavior in continuous time and, you know, to have

the discrete time step to arrive at different, I would say, you know, different dates after different duration within the continuous time model.

So, you know, when you reach, you know, a criterion in the continuous time model, you get to a next step at the top level.

And depending on the steps, you know, you can reach that earlier or later.

But when you reach this criterion, you jump, you know, at the top level.

So you make your decision and you jump at the top level.

Yes.

So one interesting question is, how could that logic be encoded into the generative model?

Another way to view that setting would be you'd have a generative model that does have like a real regular chronology, but then you're modeling samples, potentially like Poisson distributed samples from a regular continuous or discrete timeline.

So the delta T of the samples is not necessarily homogenous, but the underlying substrate of the model would be.

So observations are not coming in regularly.

Actions are potentially not being taken regularly, but there is a clock that has regularity.

That would be the sort of sampling based approach to dealing with this time heterogeneity.

Then the other approach would be somehow to figure out through composition of discrete and or continuous models.

Can you encode a temporal logic?

When this condition is met, then this happens.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

It's an interesting area.

And I mean, this, this folks psychological or just nested hybrid model.

I mean, this is just a hint.

Yeah.

And so it would be really awesome, I think, to see the MATLAB or the other code implementations.

So we could do, okay, now let's do this as a two by two matrix.

And this one is that spring attractor.

Okay.

Now this is a three by three matrix.

And this one's the lock of Volterra.

I think that if we could unroll those systems a little bit more clearly, I think we'd have some really illustrative dynamics.

Okay.

Awesome.

Well, next week, we will return for second discussion on chapter eight.

We'll just go straight to the questions.

And Ali, I think we'll be ready to do our chapter eight review when the time is right.

Sure.

Yeah.

All right.

Thank you all.

See you next time.

Thank you.

Goodbye.

Thank you.

Bye.

Bye.

Bye.